

RETAIL DATA INTEGRATION REPORT

Introduction

This project focuses on integrating and analyzing a retail sales dataset using Excel, SQL, and Python. The dataset contains **100,000** retail records covering sales, revenue, costs, margins, inventory, customer information, payment modes, brands, categories, cities, and sales channels.

SQL was used for database import and analysis, Excel was used for data summarization and visualization, and Python was used for data cleaning and validation.

1. SQL Analysis

PostgreSQL was used to import the retail dataset into a database and perform structured data analysis. A table named `retail_sales` was created and the dataset containing **100,000** records was successfully imported.

The following SQL operations were performed:

- Checked the total number of records.
- Calculated total revenue.
- Calculated total margin.
- Analyzed revenue by category.
- Analyzed revenue by city.
- Analyzed revenue by sales channel.
- Identified the top **5** brands by revenue.
- Analyzed revenue by payment mode.
- Identified low-stock records.
- Calculated the average customer age.

Key SQL Results:

- Total Records: **100,000**
- Total Revenue: **₹39,335,134.98**
- Low Stock Records: **1,729**

2. Excel Analysis

Microsoft Excel was used to summarize the retail data and create PivotTables and charts for better understanding of sales performance.

The following analyses were created using PivotTables:

- Category-wise Revenue Analysis
- City-wise Revenue Analysis
- Channel-wise Revenue Analysis
- Brand-wise Revenue Analysis

Charts were also created to visually represent the revenue distribution across different categories, cities, sales channels, and brands.

Key Excel Insights:

- Fruits generated the highest revenue among the categories.
- Kolkata generated the highest revenue among the cities.
- Omnichannel generated the highest revenue among the sales channels.
- ITC was the top-performing brand by revenue.

3. Python Data Cleaning and Validation

Python with the Pandas library was used to clean and validate the retail dataset. The main data-cleaning and validation tasks included:

- Handled missing values in `Customer_Age` using the median value.
- Replaced missing `Customer_Gender` values with "Unknown".
- Converted `Invoice_Date` into a proper datetime format.
- Checked for duplicate records.
- Verified the Revenue calculation using **Units × Selling Price**.
- Identified records where **Stock On Hand ≤ Reorder Level**.
- Checked missing values after cleaning.
- Exported the cleaned dataset as `retail_cleaned.csv`.

Python Validation Results:

- Duplicate Records: **0**
- Low Stock Records: **1,729**
- Missing Values After Cleaning: **0**
- Revenue calculation was successfully validated.

4. Key Insights and Conclusion

The analysis provided several useful insights into the retail business performance:

- The dataset contains **100,000 retail transactions**, providing a strong base for analysis.
- Total revenue generated from the dataset is approximately **₹3.93 crore**.
- **Fruits** was the highest-revenue category.
- **Kolkata** was the highest-revenue city.
- **Omnichannel** was the highest-revenue sales channel.
- **ITC** was the top-performing brand by revenue.
- **1,729 records** were identified as low-stock records and may require inventory attention.
- Python cleaning reduced the dataset's missing values to **zero** and confirmed that there were no duplicate records.

Conclusion

This project demonstrates how Excel, SQL, and Python can be combined to perform an end-to-end retail data analysis. SQL helped in storing and querying the data, Excel provided easy-to-understand summaries and visualizations, and Python helped clean and validate the dataset. Using all three tools together provided reliable insights into sales performance, customer data, and inventory conditions.